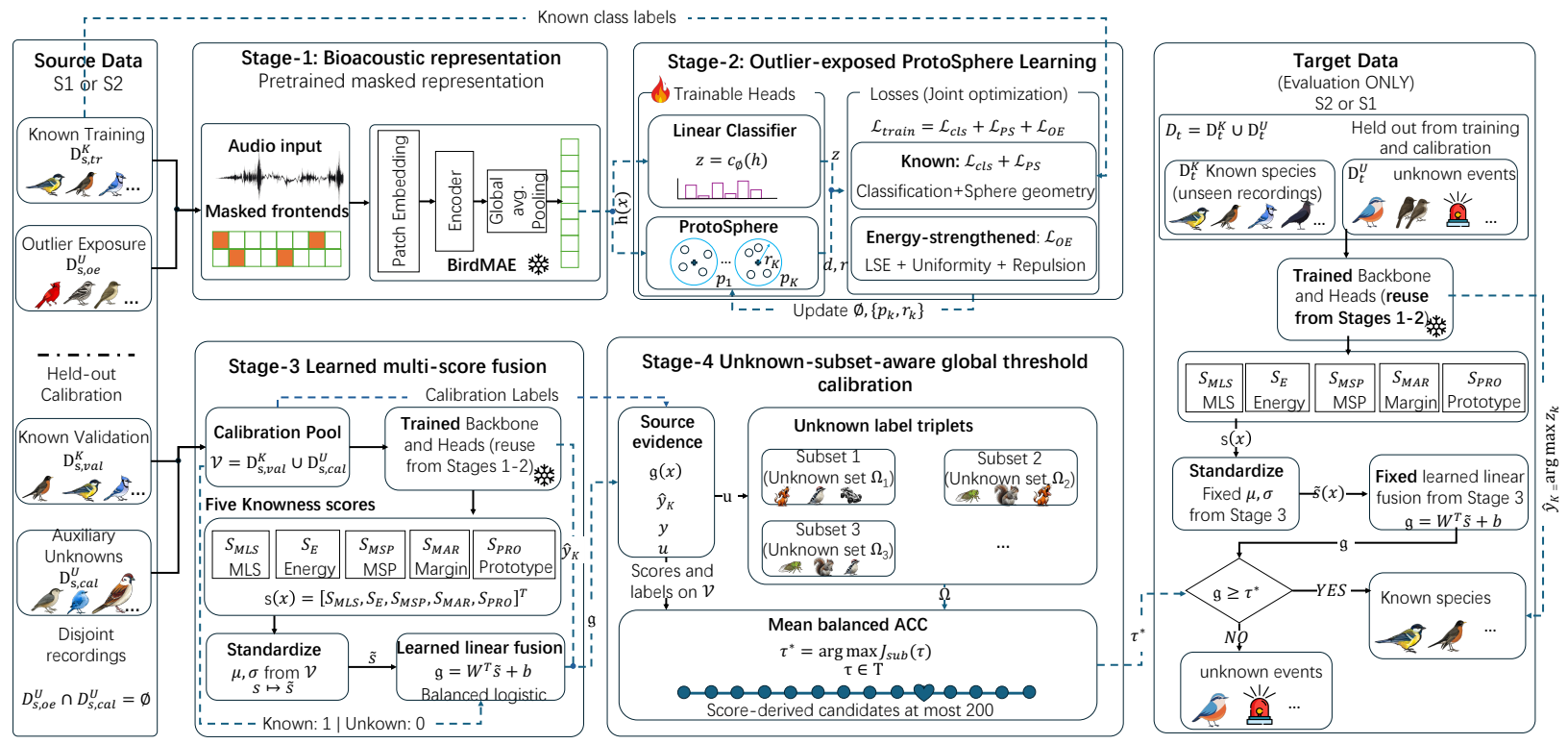

Known class labels
Source Data
S1 or S2
Known Training
$D^K_{s,tr}$
Outlier Exposure
$D^U_{s,oe}$
Held-out
Calibration
Known Validation
$D^K_{s,val}$
Auxiliary
Unknowns
$D^U_{s,cal}$
Disjoint
recordings
$D^U_{s,oe} \cap D^U_{s,cal} = \emptyset$
Stage-1: Bioacoustic representation
Pretrained masked representation
Audio input
Masked frontends
Patch Embedding
Encoder
Global avg. Pooling
BirdMAE
h(x)
Stage-2: Outlier-exposed ProtoSphere Learning
Trainable Heads
Linear Classifier
$z = c_\phi(h)$
ProtoSphere
p_1
p_K
r_K
z
d, r
Losses (Joint optimization)
$\mathcal{L}_{train} = \mathcal{L}_{cls} + \mathcal{L}_{PS} + \mathcal{L}_{OE}$
Known: $\mathcal{L}_{cls} + \mathcal{L}_{PS}$
Classification+Sphere geometry
Energy-strengthened: $\mathcal{L}_{OE}$
LSE + Uniformity + Repulsion
Update $\emptyset, \{p_k, r_k\}$
Target Data
(Evaluation ONLY)
S2 or S1
$D_t = D^K_t \cup D^U_t$
Held out from training and calibration
D^K_t Known species (unseen recordings)
D^U_t unknown events
Trained Backbone and Heads (reuse from Stages 1-2)
S_{MLS} MLS
S_E Energy
S_{MSP} MSP
S_{MAR} Margin
S_{PRO} Prototype
s(x)
Standardize
Fixed μ, σ from Stage 3
$\tilde{s}(x)$
Fixed learned linear fusion from Stage 3
$\mathfrak{g} = W^T\tilde{s} + b$
$\mathfrak{g} \geq \tau^*$
YES
NO
Known species
unknown events
$\hat{y}_K = \arg\max z_k$
Stage-3 Learned multi-score fusion
Calibration Labels
Calibration Pool
$\mathcal{V} = D^K_{s,val} \cup D^U_{s,cal}$
Trained Backbone and Heads (reuse from Stages 1-2)
Five Knowness scores
S_{MLS} MLS
S_E Energy
S_{MSP} MSP
S_{MAR} Margin
S_{PRO} Prototype
$s(x) = [S_{MLS}, S_E, S_{MSP}, S_{MAR}, S_{PRO}]^T$
$\hat{y}_K$
Standardize
μ, σ from $\mathcal{V}$
$s \mapsto \tilde{s}$
$\tilde{s}$
Learned linear fusion
$\mathfrak{g} = W^T\tilde{s} + b$
Balanced logistic
$\mathfrak{g}$
Known: 1 | Unkown: 0
Stage-4 Unknown-subset-aware global threshold calibration
Source evidence
$\mathfrak{g}(x)$
$\hat{y}_K$
y
u
Scores and labels on $\mathcal{V}$
u
Unknown label triplets
Subset 1 (Unknown set Ω_1)
Subset 2 (Unknown set Ω_2)
Subset 3 (Unknown set Ω_3)
...
Ω
Mean balanced ACC
$\tau^* = \arg\max_{\tau \in T} J_{sub}(\tau)$
Score-derived candidates at most 200
τ^*